MATH2601 – Higher Linear Algebra

Topic 3 – Linear transformations
Lecture 3.2 – Linear transformations as matrices

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Matrix multiplication as a linear map

Recall that for any matrix $A \in M_{p,q}(\mathbb{F})$, the mapping $T_A : \mathbb{F}^q \to \mathbb{F}^p$ given by $T_A(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{F}^q$ is a linear transformation.

We can use this correspondence to translate properties of linear transformations to properties of matrices. In particular, we can define $\ker(A) = \ker(T_A)$ and $\mathrm{im}(A) = \mathrm{im}(T_A)$, as well as $\mathrm{nullity}(A) = \mathrm{nullity}(T_A)$ and $\mathrm{rank}(A) = \mathrm{rank}(T_A)$.

Definition. The column space of a matrix $A \in M_{p,q}(\mathbb{F})$ is the space spanned by the columns of A as vectors in $\mathbb{F}^p$ (denoted $\mathrm{col}(A)$), and the row space is the space spanned by the rows of A as vectors in $\mathbb{F}^q$ (denoted $\mathrm{row}(A)$).

Notice that if we write $A = \begin{pmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \dots & \mathbf{c}_q \end{pmatrix}$ where each $\mathbf{c}_i \in \mathbb{F}^p$ is the ith column of A, then we have

$$\begin{aligned} \mathrm{im}(A) &= \{A\mathbf{v} : \mathbf{v} \in \mathbb{F}^q\} \\ &= \{v_1\mathbf{c}_1 + v_2\mathbf{c}_2 + \cdots + v_q\mathbf{c}_q : v \in \mathbb{F}\} \\ &= \mathrm{span}(\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_q\}) \\ &= \mathrm{col}(A). \end{aligned}$$

So $\mathrm{im}(A)$ is the span of the columns of A.

Fact. For any matrix A, we have $\dim(\mathrm{row}(A)) = \dim(\mathrm{col}(A)) = \mathrm{rank}(A)$.

Matrix of a linear map

Theorem. Suppose $(V, \mathbb{F})$ is a vector space with basis B and finite dimension q, and $(W, \mathbb{F})$ is a vector space with basis C and finite dimension p. If $T : V \to W$ is a linear transformation, then there exists a unique matrix $A \in M_{p,q}(\mathbb{F})$ such that $A[\mathbf{v}]_B = [T(\mathbf{v})]_C$ for all $\mathbf{v} \in V$.

Proof. Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_q\}$. Consider the matrix

$$A = \begin{pmatrix} [T(\mathbf{v}_1)]_C & [T(\mathbf{v}_2)]_C & \dots & [T(\mathbf{v}_q)]_C \end{pmatrix}$$

whose columns are constructed by applying T to each basis B vector and finding the result's coordinate vector over C (so the (i, j)th entry of A is the ith coordinate of $T(\mathbf{v}_j)$ over C).

For any $\mathbf{v} \in V$, write $\mathbf{v} = \sum \lambda_i \mathbf{v}_i$ for some $\lambda_i \in \mathbb{F}$. Then since T and the act of taking coordinates are linear transformations,

$$A[\mathbf{v}]_B = A(\lambda_i)_{1 \le i \le q} = \sum_{i=1}^{q} \lambda_i [T(\mathbf{v}_i)]_C = \left[T\left(\sum_{i=1}^{q} \lambda_i \mathbf{v}_i \right) \right]_C = [T(\mathbf{v})]_C.$$

To see that the matrix A is unique, note that if there were some other matrix A' satisfying the requirements, then $A[\mathbf{v}]_B = A'[\mathbf{v}]_B$ for all $\mathbf{v} \in V$, meaning $(A - A')\mathbf{u} = \mathbf{0}$ for all $\mathbf{u} \in \mathbb{F}^q$, so $A - A' = \mathbf{0}$ and thus $A = A'$.

Matrix of a linear map

Corollary. Any matrix $A \in M_{p,q}(\mathbb{F})$ determines a unique linear map $T : V \to W$ given by $[T(\mathbf{v})]_C = A[\mathbf{v}]_B$ for all $\mathbf{v} \in V$.

Corollary. Given finite-dimensional vector spaces $(V, \mathbb{F})$ and $(W, \mathbb{F})$ with $\dim(V) = q$ and $\dim(W) = p$, we have $\mathcal{L}(V, W) \cong M_{p,q}(\mathbb{F})$.

Definition. Given finite-dimensional vector spaces $(V, \mathbb{F})$ and $(W, \mathbb{F})$ with bases B and C respectively, and a linear transformation $T : V \to W$, the tarnsformation matrix of T with respect to B and C is denoted by $[T]_C^B$ and is the $\dim(W) \times \dim(V)$ matrix over $\mathbb{F}$ given by

$$[T]_C^B = \begin{pmatrix} [T(\mathbf{v}_1)]_C & [T(\mathbf{v}_2)]_C & \dots & [T(\mathbf{v}_q)]_C \end{pmatrix},$$

with the property that $[T(\mathbf{v})]_C = [T]_C^B[\mathbf{v}]_B$ for all $\mathbf{v} \in V$.

Definition. Given a vector space V with bases B and C, the transformation matrix $[\mathrm{id}]_C^B$ is called the change of basis matrix from B to C.

In particular, for all $\mathbf{v} \in V$ we have $[\mathbf{v}]_C = [\mathrm{id}]_C^B[\mathbf{v}]_B$.

Example – Identity map, nonstandard to standard bases

Exercise. Find the transformation matrix for the map $\mathrm{id} : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ with respect to the basis $B = \left\{2t^2 - t + 1, t^2 - t, 2t^2 - t + 2\right\}$ for the domain and the standard basis $\mathcal{S} = \{1, t, t^2\}$ for the codomain.

Solution. We want to find $[\mathrm{id}]_{\mathcal{S}}^{B}$ which is the matrix with columns given by $[\mathrm{id}(p)]_{\mathcal{S}}$ for each $p \in B$.

Here $\mathrm{id}(2t^2 - t + 1) = (1)1 + (-1)t + (2)t^2$, so the corresponding coordinate vector is $[\mathrm{id}(2t^2 - t + 1)]_{\mathcal{S}} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$.

Similarly $[\mathrm{id}(t^2 - t)]_{\mathcal{S}} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$ and $[\mathrm{id}(2t^2 - t + 2)]_{\mathcal{S}} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

Since each of these vectors is already their coordinate vector over the standard basis $\mathcal{S}$, we have

$$[\mathrm{id}]_{\mathcal{S}}^{B} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & -1 & -1 \\ 2 & 1 & 2 \end{pmatrix}.$$

Example – Identity map, standard to nonstandard bases

Exercise. Find the transformation matrix for the map $\mathrm{id} : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ with respect to the standard basis $\mathcal{S} = \{1, t, t^2\}$ for the domain and the basis $B = \left\{2t^2 - t + 1, t^2 - t, 2t^2 - t + 2\right\}$ for the codomain.

Solution. We want to find $[\mathrm{id}]_B^{\mathcal{S}}$ which is the matrix with columns given by $[\mathrm{id}(p)]_B$ for each $p \in \mathcal{S}$. To find $[\mathrm{id}(1)]_B$ we need to solve the equation $\mathrm{id}(1) = 1 = \lambda_1(2t^2 - t + 1) + \lambda_2(t^2 - t) + \lambda_3(2t^2 - t + 2)$ for $\lambda_i \in \mathbb{R}$, which is a system of equations with corresponding matrix $\left(\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ -1 & -1 & -1 & 0 \\ 2 & 1 & 2 & 0 \end{array}\right)$.

For $[\mathrm{id}(t)]_B$ and $[\mathrm{id}(t^2)]_B$ we need to solve similar systems, except with the last column being replaced by $\mathbf{e}_2$ and $\mathbf{e}_3$ respectively. We can solve all three systems simultaneously by row-reducing the augmented matrix

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ -1 & -1 & -1 & 0 & 1 & 0 \\ 2 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 2 & 2 \\ 0 & 1 & 0 & 0 & -2 & -1 \\ 0 & 0 & 1 & 1 & -1 & -1 \end{array}\right).$$

The columns to the right of the bar are precisely the coordinate vectors, so

$$[\mathrm{id}]_B^{\mathcal{S}} = \begin{pmatrix} -1 & 2 & 2 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{pmatrix}.$$

Transformation matrix inverses

Notice that in the previous example, the way $[\mathrm{id}]_B^{\mathcal{S}}$ was constructed is the same method as is used to produce the inverse of a matrix. Indeed, we can check that $[\mathrm{id}]_B^{\mathcal{S}} = ([\mathrm{id}]_{\mathcal{S}}^B)^{-1}$.

Lemma. Given a vector space V with bases B and C, the change of basis matrix from C to B is the inverse of the change of basis matrix from B to C, that is, $[\mathrm{id}]_B^C = ([\mathrm{id}]_C^B)^{-1}$.

Proof. For any $\mathbf{v} \in V$, we have

$$[\mathbf{v}]_C = [\mathrm{id}]_C^B[\mathbf{v}]_B = [\mathrm{id}]_C^B([\mathrm{id}]_B^C[\mathbf{v}]_C) = ([\mathrm{id}]_C^B[\mathrm{id}]_B^C)[\mathbf{v}]_C.$$

As this is true for all $\mathbf{v} \in V$, we have $[\mathrm{id}]_C^B[\mathrm{id}]_B^C = I$, so $[\mathrm{id}]_B^C = ([\mathrm{id}]_C^B)^{-1}$.

More generally we have the following (proven after the next theorem).

Lemma. Given a linear transformation $T : V \to W$ for vector spaces V and W with respective bases B and C, if T is invertible then $[T^{-1}]_B^C = ([T]_C^B)^{-1}$.

Corollary. Given any transformation matrix A for a linear map T, the matrix A is invertible if and only if the function T is invertible.

Corollary. The space of bijective linear maps from vector spaces $(V, \mathbb{F})$ to $(W, \mathbb{F})$ with dimension n is isomorphic to $\mathrm{GL}(n, \mathbb{F})$.

Example – Identity map, nonstandard to nonstandard bases

Exercise. Find the transformation matrix for the map $\mathrm{id} : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ with respect to the basis $C = \{t^2 - 1, 2t^2 - t - 2, 2t^2\}$ for the domain and the basis $B = \{2t^2 - t + 1, t^2 - t, 2t^2 - t + 2\}$ for the codomain.

Solution. We want to find $[\mathrm{id}]_B^C$ which is the matrix with columns given by $[\mathrm{id}(p)]_B$ for each $p \in C$. To find $[\mathrm{id}(t^2 - 1)]_B$ we need to solve the equation

$$\lambda_1(2t^2 - t + 1) + \lambda_2(t^2 - t) + \lambda_3(2t^2 - t + 2) = t^2 - 1$$

for $\lambda_i \in \mathbb{R}$, and likewise for $[\mathrm{id}(2t^2 - t - 2)]_B$ and $[\mathrm{id}(2t^2)]_B$. We can solve all three implied systems of linear equations simultaneously by row-reducing the augmented matrix

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & -2 & 0 \\ -1 & -1 & -1 & 0 & -1 & 0 \\ 2 & 1 & 2 & 1 & 2 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 4 & 4 \\ 0 & 1 & 0 & -1 & 0 & -2 \\ 0 & 0 & 1 & -2 & -3 & -2 \end{array}\right).$$

The columns to the right of the bar are precisely the coordinate vectors, so

$$[\mathrm{id}]_B^C = \begin{pmatrix} 3 & 4 & 4 \\ -1 & 0 & -2 \\ -2 & -3 & -2 \end{pmatrix}.$$

Transformation matrix multiplication

Notice that in the prior example, the result $[\mathrm{id}]_B^C$ is equivalent to the matrix product $[\mathrm{id}]_B^{\mathcal{S}}[\mathrm{id}]_{\mathcal{S}}^C$ where $\mathcal{S} = \{1, t, t^2\}$ is the standard basis for $\mathcal{P}_2(\mathbb{R})$.

A more general result related to this is given below.

Lemma. Given two linear transformations $S : U \to V$ and $T : V \to W$ for vector spaces U, V, and W with bases B, C, and D respectively, we have $[T \circ S]_D^B = [T]_D^C[S]_C^B$.

Proof. Note that for all $\mathbf{u} \in U$,

$$\begin{aligned}
[T \circ S]_D^B[\mathbf{u}]_B &= [T \circ S(\mathbf{u})]_D \\
&= [T(S(\mathbf{u}))]_D \\
&= [T]_D^C[S(\mathbf{u})]_C \\
&= [T]_D^C[S]_C^B[\mathbf{u}]_B.
\end{aligned}$$

Since this is true for all $\mathbf{u} \in U$, we must have $[T \circ S]_D^B = [T]_D^C[S]_C^B$.

Corollary. For invertible T, we find $[T^{-1}]_B^C[T]_C^B = [T^{-1} \circ T]_B^B = [\mathrm{id}]_B^B = I$. This proves the previous result $[T^{-1}]_B^C = ([T]_C^B)^{-1}$ for any invertible T.

Example – Arbitrary map, standard to standard bases

Exercise. Find the transformation matrix for the map $T : \mathcal{P}_2(\mathbb{R}) \to \mathbb{R}^3$ given by $T(p) = \begin{pmatrix} p(0) \\ p(1) \\ p(2) \end{pmatrix}$ (with respect to the standard bases).

Solution. Writing $\mathcal{S} = \{1, t, t^2\}$ for the standard basis of $\mathcal{P}_2(\mathbb{R})$ and $\mathcal{S}' = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ for the standard basis of $\mathbb{R}^3$, we want to find $[T]^{\mathcal{S}}_{\mathcal{S}'}$ which is the matrix with columns given by $[T(p)]_{\mathcal{S}'}$ for each $p \in \mathcal{S}$.

Since $T(1) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $T(t) = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$, and $T(t^2) = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}$, and each of these vectors is already their coordinate vector over the standard basis $\mathcal{S}'$, we have

$$[T]^{\mathcal{S}}_{\mathcal{S}'} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}.$$

Example – Arbitrary map, nonstandard to standard bases

Exercise. Find the transformation matrix for $T : \mathcal{P}_2(\mathbb{R}) \to \mathbb{R}^3$ given by $T(p) = \begin{pmatrix} p(0) \\ p(1) \\ p(2) \end{pmatrix}$ with respect to the basis $B = \{t-1, t^2-2, t^2-t\}$ for $\mathcal{P}_2(\mathbb{R})$ and the standard basis $\mathcal{S}'$ for $\mathbb{R}^3$.

Solution. Here $T(t-1) = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, $T(t^2-2) = \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$, $T(t^2-t) = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$, and each of these vectors is already their coordinate vector over the standard basis $\mathcal{S}'$, so

$$[T]^B_{\mathcal{S}'} = \begin{pmatrix} -1 & -2 & 0 \\ 0 & -1 & 0 \\ 1 & 2 & 2 \end{pmatrix}.$$

Alternatively, writing $\mathcal{S} = \{1, t, t^2\}$ for the standard basis for $\mathcal{P}_2(\mathbb{R})$,

$$[T]^B_{\mathcal{S}'} = [T]^{\mathcal{S}}_{\mathcal{S}'}[\mathrm{id}]^B_{\mathcal{S}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix} \begin{pmatrix} -1 & -2 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -2 & 0 \\ 0 & -1 & 0 \\ 1 & 2 & 2 \end{pmatrix}.$$

Example – Arbitrary map, nonstandard to nonstandard

Exercise. Find the transformation matrix for $T : \mathcal{P}_2(\mathbb{R}) \to \mathbb{R}^3$ given by $T(p) = \begin{pmatrix} p(0) \\ p(1) \\ p(2) \end{pmatrix}$ with respect to the basis $B = \{t-1, t^2-2, t^2-t\}$ for $\mathcal{P}_2(\mathbb{R})$ and the basis $C = \left\{ \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \right\}$ for $\mathbb{R}^3$.

Solution. From the previous example, $[T]^B_{\mathcal{S}'} = \begin{pmatrix} -1 & -2 & 0 \\ 0 & -1 & 0 \\ 1 & 2 & 2 \end{pmatrix}$ where $\mathcal{S}'$ is the standard basis for $\mathbb{R}^3$. Now we want each column's coordinates over the basis C. As before, we want to take the right half of the row-reduced matrix

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & -2 & 0 \\ -1 & -1 & -1 & 0 & -1 & 0 \\ 2 & 1 & 2 & 1 & 2 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 4 & 4 \\ 0 & 1 & 0 & -1 & 0 & -2 \\ 0 & 0 & 1 & -2 & -3 & -2 \end{array}\right).$$

Alternatively,

$$[T]^B_C = [\mathrm{id}]^{\mathcal{S}'}_C [T]^B_{\mathcal{S}'} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & -1 & -1 \\ 2 & 1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} -1 & -2 & 0 \\ 0 & -1 & 0 \\ 1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 4 & 4 \\ -1 & 0 & -2 \\ -2 & -3 & -2 \end{pmatrix}.$$

Change of basis formula

The previous examples and theorems combine to give the following formula.

Theorem. (Change of basis formula)
Given a linear transformation $T : V \to W$ between vector spaces V and W with respective standard bases $\mathcal{S}$ and $\mathcal{S}'$ and given bases B and C, the transformation matrix of T with respect to B and C is given by the formula

$$[T]_C^B = [\mathrm{id}_W]_C^{\mathcal{S}'}[T]_{\mathcal{S}'}^{\mathcal{S}}[\mathrm{id}_V]_{\mathcal{S}}^B.$$

This is sometimes represented by a so-called commutative diagram:

$$\begin{tikzcd}
\text{(basis } B\text{)}\ V \arrow[r, "{[T]_C^B}"] \arrow[d, "{[\mathrm{id}_V]_{\mathcal{S}}^B}"'] & W\ \text{(basis } C\text{)} \arrow[d, "{[\mathrm{id}_W]_{\mathcal{S}'}^C}"] \\
\text{(basis } \mathcal{S}\text{)}\ V \arrow[r, "{[T]_{\mathcal{S}'}^{\mathcal{S}}}"] & W\ \text{(basis } \mathcal{S}'\text{)}
\end{tikzcd}$$

Notice that while $[T]_C^B$ might be hard to find in general, the matrices $[\mathrm{id}_V]_{\mathcal{S}}^B$ and $[T]_{\mathcal{S}'}^{\mathcal{S}}$ are relatively easy to find, as is $[\mathrm{id}_W]_{\mathcal{S}'}^C$ whose inverse is $[\mathrm{id}_W]_C^{\mathcal{S}'}$.

Sometimes it is even easier to find $[T]_{\mathcal{S}'}^B$ (like in the previous example), in which case we can just use the formula $[T]_C^B = [\mathrm{id}_W]_C^{\mathcal{S}'}[T]_{\mathcal{S}'}^B$.

Example – Change of basis formula

Exercise. Find the transformation matrix for the map $T : \mathbb{R}^3 \to \mathbb{R}^2$ given by $T(\mathbf{v}) = \begin{pmatrix} v_1 - v_3 \\ 2v_1 + v_2 \end{pmatrix}$ where $\mathbf{v} = (\, v_1 \;\, v_2 \;\, v_3 \,)^T$ with respect to the bases

$$B = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} \text{ for } \mathbb{R}^3 \text{ and } C = \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \begin{pmatrix} -1 \\ 3 \end{pmatrix} \right\} \text{ for } \mathbb{R}^2.$$

Solution. We want to find $[T]_C^B = [\mathrm{id}]_C^{\mathcal{S}'} [T]_{\mathcal{S}'}^{\mathcal{S}} [\mathrm{id}]_{\mathcal{S}}^B$ where

$[\mathrm{id}]_{\mathcal{S}}^B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$, $[\mathrm{id}]_{\mathcal{S}'}^C = \begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix}$, and $[T]_{\mathcal{S}'}^{\mathcal{S}} = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix}$.

Here $[T]_{\mathcal{S}'}^{\mathcal{S}} [\mathrm{id}]_{\mathcal{S}}^B = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 2 & 3 \end{pmatrix}$ and $[\mathrm{id}]_C^{\mathcal{S}'} = \left([\mathrm{id}]_{\mathcal{S}'}^C \right)^{-1} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$, thus

$$[T]_C^B = [\mathrm{id}]_C^{\mathcal{S}'} [T]_{\mathcal{S}'}^{\mathcal{S}} [\mathrm{id}]_{\mathcal{S}}^B = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 3 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 6 & 2 & 3 \\ 5 & 2 & 3 \end{pmatrix}.$$

Alternatively, take the right half of the row-reduced matrix

$$\left(\begin{array}{cc|ccc} 1 & -1 & 1 & 0 & 0 \\ -2 & 3 & 3 & 2 & 3 \end{array} \right) \sim \left(\begin{array}{cc|ccc} 1 & 0 & 6 & 2 & 3 \\ 0 & 1 & 5 & 2 & 3 \end{array} \right).$$